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Next item →

Instructions

1. What is the primary principle behind the k-Nearest Neighbors (kNN) algorithm? **1 / 1 point**

- (●) It classifies a data point based on the majority class of its nearest neighbors.
- ( ) It divides the dataset into k clusters and predicts the class based on these clusters.
- ( ) It uses a fixed number of nearest neighbors to determine the decision boundary.
- ( ) It predicts the value of a data point based on the average of all data points.

Correct
Correct! kNN classifies a data point based on how similar points (neighbors) are classified.

2. In kNN, how does the choice of 'k' affect the model? **1 / 1 point**

- ( ) A smaller 'k' increases computational complexity.
- (●) A smaller 'k' can make the model more sensitive to noise, while a larger 'k' makes it more resistant.
- ( ) A larger 'k' significantly reduces the training time.

Correct
Correct! Smaller 'k' values are more sensitive to noise, larger values are more stable.

3. What does a kernel smoother do in the context of data analysis? **1 / 1 point**

- ( ) It categorizes data into distinct groups.
- (●) It estimates a smooth curve or surface through the data points.
- ( ) It amplifies the signal in noisy data.
- ( ) It separates data into linearly separable classes.

Correct
Correct! Kernel smoothers are used for estimating a smooth function through data.

Kernel Smoothers Quiz

Review Learning Objectives

coach

Ready to review what you've learned before starting the assignment? I'm here to help.

Help me practice Let's chat

Assignment details

Submitted Nov 16, 2:45 AM IST **Attempts** Unlimited **Try again**

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